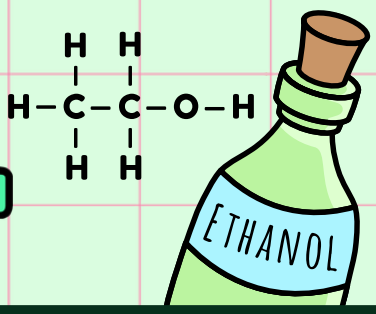
ETHANOL

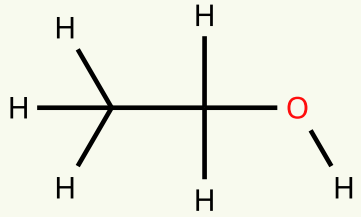
A Simple Alcohol with Big Uses



Compound Identity

Common name	**: Ethanol**
IUPAC name	**: Ethanol**
Molecular formula	**: C_2H_6O**
molecular weight	**: 46.07 g/mol**

Structure



Structural formula (displayed)

Functional Groups

- **Hydroxyl group (-OH)**

Notable feature

- **Two carbon atoms (a short-chain alcohol).**

Important Properties

Appearance / State
Colorless, clear liquid

Solubility
Miscible with water and most organic solvents.

Melting / Boiling Point
Melting point: -114.1 °C
Boiling point: 78.4 °C

Polarity
Polar (due to -OH group).

Flammability
Highly flammable.

What is Amazing About It?

What is Amazing About This Compound?

Ethanol is a versatile organic compound that is commonly used as a solvent and fuel. Its polar –OH group and nonpolar carbon chain allow it to interact with different substances, making it useful in many applications.

Uses / Value

Everyday uses
- **Beverages (alcoholic drinks)**
- **Solvent (paints, perfumes, medicines)**

Medical uses
- **Antiseptic (hand sanitizers)**
- **Disinfectant**

Industrial uses
- **Fuel (bioethanol, cleaner energy)**
- **Chemical feedstock**

Chemical engineering connec
Used in fermentation, distillation, and as a solvent in many chemical processes.
Also important in the production of fuels, plastics, and other chemicals.

Risks & Safety

Health risks
- **Highly flammable (fire hazard).**
- **Can cause dizziness, loss of coordination, and long-term health effects with excessive use**
- **Keep away from open flames and heat sources.**

Fire / Reactivity
- **Vapors can form explosive mixtures with air.**
- **Handle with care and use proper ventilation.**

Environmental concerns
- **Can contribute to air pollution if released in large amounts.**

References

- **National Center for Biotechnology Information. (2025). PubChem compound summary for CID 702, Ethanol.**
- **-National Center for Biotechnology Information. (n.d.). Ethanol. PubChem. https://pubchem.ncbi.nlm.nih.gov/compound/Ethanol**

Clean energy for a brighter future